

FPGA-based Ethernet audio/video bridging network implementation

Xilinx has demonstrated a field-programmable gate array (FPGA)-based Ethernet audio/video bridging (EAVB) network implementation optimised for carrying high-speed data traffic within the automobile.

The EAVB development kit is based on the Xilinx Automotive (XA) Spartan(R)-6 FPGA, which provides system integration capabilities and low total cost for high-volume automotive applications. The solution offers a complete network interface solution, while providing flexibility for addressing a customer's unique application interface needs. It also offers maximum integration potential where Xilinx FPGAs are already commonly used.

In the demonstration, video data from four different cameras was streamed over an EAVB network to an FPGA where further driver-assistance video and graphics processing could be performed. In addition, separate infotainment audio/video source streaming to a remote display was performed simultaneously on the same network. Low-latency MJPEG video compression was implemented to further manage the network bandwidth.

Industry's first integrated MoCA 2.0 SoC portfolio

Broadcom claims to have introduced the industry's first Multimedia over Coax Alliance (MoCA) 2.0-integrated portfolio including six new set-top box (STB) and hybrid Internet protocol (IP) gateway system-on-chip (SoC) platforms. Integrating MoCA 2.0 directly in the STB SoC, the portfolio realises the MoCA 2.0 benefits including double the video bandwidth, increased security and lower power.

BCM7425 dual high-definition (HD) transcoding gateway STB SoC features dual HD decoding and dual transcoding support for streaming si-



NEX Robotics' autonomous transport vehicle

multaneous video broadcast content wirelessly to multiple devices. Broadcom's 40nm MoCA 2.0-integrated STB and gateway platforms are powered by its dual-thread Zephyr CPU that delivers 3000 Dhrystone MIPS.

MOSFET drivers for telecom and server power supplies

Texas Instruments claims to have introduced the industry's first 120V rated, 4A high- and low-side drivers (TTL- and CMOS-compatible input versions) that drive both n-channel high-side and low-side field-effect transistors. The combined features improve efficiency and enhance reliability in telecom, server and industrial power supply designs with 100V surge requirements.

Pull-up and pull-down resistance of 0.9 ohm minimises switching loss as the MOSFET transitions through the Miller Plateau. Inputs support -10V of direct current, and allow direct interface to gate-driver transformers without the need for rectification diodes.

Intelligent autonomous transport vehicle

NEX Robotics has designed a four-seater electric-powered vehicle that can manoeuvre autonomously based on

the fixed waypoints. It can be driven manually as well.

For autonomous navigation, the vehicle primarily uses GPS, inertial guidance and fluxgate magnetometers. The on-board computer estimates the vehicle's lateral and longitudinal position with the help of sensors. Any undesired deviation from the path is corrected in real time by a steer-by-wire system.

For safety considerations, the top speed of the vehicle is limited to 20 kmph in the autonomous mode. Ultrasonic obstacle sensors are used for collision avoidance.

Solar-powered pesticide sprayer

David Raja Beleau, assistant director of Horticulture Kadayam, Tamil Nadu, has developed a solar-powered battery-operated sprayer.

The sprayer is fitted with an electric motor operated on 12V and 7-ampere rechargeable battery. It can hold 16 litres of pesticide solution. Once fully charged, it can be used for eight hours continuously without the need of any fuel like petrol or diesel to operate it.

Its battery can be charged by solar energy via the solar power system attached to it. The solar panel of the sprayer is attached to a helmet and connected to the battery unit of the sprayer.

Check efytimes.com for more news, daily